

# **MDS-UPDRS Finger Tapping Score**

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

**Instructions to the examiner:**

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

**Scoring Criteria:**

**0 (Normal):** No problems

**1 (Slight):** Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

**2 (Mild):** Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

**3 (Moderate):** Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

**4 (Severe):** Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

## ## EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [222, 192, 223, 159, 167, 219, 141, 189, 155, 224]

Tap amplitudes (pixels): [178.21896644296868, 168.9644933114647, 174.41330224498358, 159.84992962150469, 170.66048165876012, 168.58232410309213, 160.5615146914104, 152.620444239951, 162.78820596099706, 158.14234094637652, 154.38264151127873]

---

## Gemini 3 Pro

Based on the extracted features provided, here is the analysis according to the MDS-UPDRS Item 3.4 criteria:

### 1. Amplitude Analysis:

- **Start:** The first tap has an amplitude of approximately **178 pixels**.
- **Middle:** By tap 5, the amplitude is **170 pixels**. It remains relatively stable through the first half of the sequence.
- **End:** The amplitude drops to **152 pixels** at tap 8 and ends at **154 pixels** at tap 11.
- **Conclusion:** There is a decrement in amplitude, but it becomes most noticeable towards the final few taps (taps 8-11). This fits the criteria for **Score 1** ("Amplitude decrements near the end of the 10 taps") rather than Score 2 ("decrements midway").

### 2. Rhythm and Speed Analysis:

- **Variability:** The tap periods (time between taps) show significant irregularity: [222, 192, 223, 159, 167, 219, 141, 189, 155, 224].
- **Hesitations:** The subject fluctuates between faster taps (~140-160 frames) and slower taps (~220 frames). This indicates a rhythm that is broken by hesitations, but there are no distinct "arrests" or "freezes" (which would typically be much longer durations).
- **Conclusion:** The irregular rhythm fits the criteria for **Score 1** ("The regular rhythm is broken with one or two interruptions or hesitations"). While the rhythm is quite variable, the lack of severe slowing or mid-sequence amplitude crash keeps it out of the higher impairment categories.

### Final Scoring:

The performance exhibits a regular rhythm broken by hesitations and a slight decrement in amplitude near the end of the task. This aligns with the criteria for **Slight** impairment.

